Lecture 2

Ordinary differential equations: the mechanics contract

Objectives. *A differential equation determines a motion from its initial state when its right-hand side has sufficient regularity. We construct the motion by contraction. We then identify its maximal time interval and prove that a finite endpoint forces escape from every compact subset of the state space. These solutions define the flow used in mechanics.*

By the end of this lecture you will be able to:

- Construct a local solution for a continuously differentiable vector field.
- Bound changes in a solution caused by changes in its initial state.
- Use compact escape to prove existence for all time.
- State the domain and group law of a flow, and its variational equation.

2.1 The integral equation

Let $m \in \mathbb{N}$ and let $U \subset \mathbb{R}^m$ be open. A *vector field* on U assigns a vector $F(x) \in \mathbb{R}^m$ to each state $x \in U$. The equation is *autonomous* when F does not depend on time. A *solution through* $z \in U$ is a continuously differentiable map $x : J \to U$, on an open interval J containing zero, such that

$$\dot{x}(t) = F(x(t)), \qquad x(0) = z.$$

Throughout this lecture, $F \in C^1(U, \mathbb{R}^m)$ unless another hypothesis is stated. The fundamental theorem of calculus gives the equivalent condition

$$x(t) = z + \int_0^t F(x(s))\, \mathrm{d}s. \tag{2.1}$$

Conversely, a continuous path satisfying this condition is continuously differentiable, because its integrand is continuous. Autonomy permits time translation. If x solves the equation on J and $s \in J$, then $t \mapsto x(t+s)$ solves it on $J - s$, since its derivative is $F(x(t+s))$.

We use the Euclidean norm on $\mathbb{R}^m$. For a real $m \times m$ matrix A, its operator norm is

$$\|A\| = \sup_{\substack{y \in \mathbb{R}^m \\ \|y\| = 1}} \|Ay\|.$$

A field is *Lipschitz* on a set $S \subset U$ if some constant $L \geq 0$ satisfies

$$\|F(x) - F(y)\| \leq L \|x - y\|, \qquad x, y \in S.$$

It is *locally Lipschitz* if every point has a neighbourhood with such a bound. A continuously differentiable field is locally Lipschitz. Write DF for the Jacobian matrix of F. On a closed

ball $\overline{B}(a,r) \subset U$, continuity bounds $\|DF\|$ by a constant L. For x and y in this ball, the chain rule along their segment gives

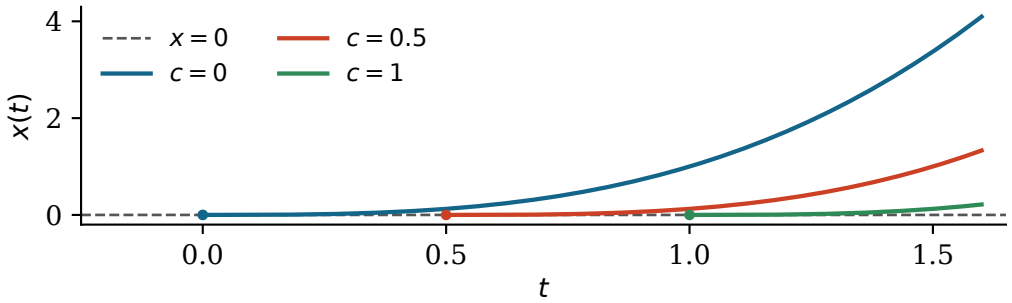
$$F(y) - F(x) = \int_0^1 DF(x + s(y-x))(y-x)\,\mathrm{d}s.$$

The segment stays in the ball, so taking norms gives the required bound.

Continuity alone does not give uniqueness. On $\mathbb{R}$, the field $F(x) = 3\,|x|^{2/3}$ has no Lipschitz bound near zero, since $|F(h) - F(0)|\,/|h| = 3\,|h|^{-1/3}$ for $h \neq 0$. For every waiting time $c \geq 0$, define

$$x_c(t) = \begin{cases} 0, & t \leq c, \\ (t-c)^3, & t > c. \end{cases}$$

Both the value and derivative match at c. After c, its derivative is $3(t-c)^2 = 3\,|x_c(t)|^{2/3}$. Before c, both sides vanish. Thus x_c and the zero solution all pass through $x(0) = 0$. The graph shows different possible motions from that initial state.



The zero solution and the solutions with waiting times 0, 1/2 and 1 coincide before their departure times. The dots mark those times.

2.2 Picard–Lindelöf by contraction

The completeness proof in Theorem 1.1 applies coordinate by coordinate to continuous $\mathbb{R}^m$-valued paths on a compact interval. The contraction principle in Theorem 1.6 gives a unique fixed point for a contraction of a nonempty closed subset of such a Banach space.

> **Theorem 2.1** (Local existence and uniqueness). *Let $F \in C^1(U, \mathbb{R}^m)$ and $a \in U$. There are $\tau > 0$ and an open neighbourhood V of a such that every $z \in V$ has a solution on $(-\tau, \tau)$. Any two solutions with the same initial state agree on the intersection of their intervals.*

Proof. Choose $r > 0$ with $\overline{B}(a,r) \subset U$. On this ball choose bounds $\|F(x)\| \leq C_0$ and $\|DF(x)\| \leq L$. Choose $\tau > 0$ so that $\tau C_0 \leq r/2$ and $\tau L < 1$. These conditions also allow $C_0 = 0$ or $L = 0$. Take $V = B(a, r/2)$. For each $z \in V$, work in the closed set

$$X = \{x \in C([-\tau,\tau], \mathbb{R}^m) : x(t) \in \overline{B}(a,r)\}, \qquad \|x\|_X = \sup_{|t| \leq \tau} \|x(t)\|.$$

This set is nonempty and closed in the complete path space. Define the Picard map $\mathcal{P}_z : X \to X$ by

$$(\mathcal{P}_z x)(t) = z + \int_0^t F(x(s))\,\mathrm{d}s.$$

It maps into X because $\|(\mathcal{P}_z x)(t) - a\| \le \|z - a\| + \tau C_0 \le r$. The Lipschitz bound gives

$$\|\mathcal{P}_z x - \mathcal{P}_z y\|_X \le \tau L \, \|x - y\|_X .$$

The contraction principle supplies a unique fixed point. By Equation (2.1), it is a solution.

Every other solution through z remains in the ball for sufficiently small times by continuity, so the same contraction argument on a shorter interval makes it agree there. If two solutions meet at another time, shift that time to zero and apply the construction with a equal to their common state. Their set of agreement is therefore open and closed in their common interval. It is nonempty, and an interval is connected. Hence they agree throughout that interval. ∎

For example, consider $\dot{x} = x^2$ near $a = 1$. On $[1/2, 3/2]$ we can take $C_0 = 9/4$ and $L = 3$. The choice $\tau = 1/10$ gives $\tau C_0 = 9/40 < 1/4$ and $\tau L = 3/10 < 1$. For initial state $z = 1$, start the iteration with $x_0(t) = 1$. Successive integration gives

$$x_1(t) = 1 + t, \qquad x_2(t) = 1 + t + t^2 + \frac{t^3}{3}.$$

The theorem proves convergence to a solution, rather than merely producing formal approximations.

The same construction also controls its initial state. For $z, w \in V$, the two fixed points satisfy

$$\|x_z - x_w\|_X \le \|z - w\| + \tau L \, \|x_z - x_w\|_X , \qquad \|x_z - x_w\|_X \le \frac{\|z - w\|}{1 - \tau L}. \tag{2.2}$$

Moreover, $\|x_z(t) - x_z(s)\| \le C_0 \, |t - s|$. Together these bounds show continuity in time and initial state on this common local interval.

2.3 Grönwall and dependence on the initial state

The local estimate has a denominator that restricts the interval. Grönwall's inequality gives a bound over longer intervals.

Lemma 2.2 (Grönwall). *Let u and b be continuous nonnegative functions on $[0, H]$, and let $a \ge 0$. If*

$$u(t) \le a + \int_0^t b(s) u(s) \, \mathrm{d}s, \qquad 0 \le t \le H,$$

then $u(t) \le a \exp(\int_0^t b(s) \, \mathrm{d}s)$ for $0 \le t \le H$.

Proof. For $\varepsilon > 0$, put $w(t) = a + \varepsilon + \int_0^t b(s) u(s) \, \mathrm{d}s$. Then $u \le w$ and $w' = bu \le bw$. The product rule gives

$$\frac{\mathrm{d}}{\mathrm{d}t} \left(w(t) e^{-\int_0^t b(s) \, \mathrm{d}s} \right) \le 0.$$

Integrating gives $u(t) \le (a + \varepsilon) e^{\int_0^t b(s) \, \mathrm{d}s}$. Let $\varepsilon \downarrow 0$. ∎

Proposition 2.3 (Separation of solutions). *Suppose F is Lipschitz with constant L on a set $S \subset U$. If two solutions x_z and x_w stay in S between times zero and t, then*

$$\|x_z(t) - x_w(t)\| \le e^{L|t|} \, \|z - w\| . \tag{2.3}$$

Proof. For $t \geq 0$, subtract the integral equations and take norms:

$$\|x_z(t) - x_w(t)\| \leq \|z - w\| + L \int_0^t \|x_z(s) - x_w(s)\| \, \mathrm{d}s.$$

Apply Lemma 2.2 with $a = \|z - w\|$ and $b = L$. For negative times, the paths $x_z(-s)$ and $x_w(-s)$ solve the equation with field $-F$, which has the same Lipschitz constant. ∎

For $F(x) = \lambda x$ on $\mathbb{R}$, with $\lambda \geq 0$, the solution is $x_z(t) = e^{\lambda t} z$. For $t \geq 0$ its separation is exactly $e^{\lambda t} |z - w|$, so the exponential bound can be attained. With $z = w$, the estimate also proves uniqueness on every interval where one Lipschitz bound applies.

2.4 Maximal solutions and compact escape

A solution is *maximal* if it has no extension to a larger open interval as a solution in U. An extension must agree on the original interval. Maximal does not mean defined for all time.

Theorem 2.4 (Maximal interval and compact escape). *For every $z \in U$, there is a unique maximal solution x_z, on an interval*

$$J_z = (T_-(z), T_+(z)), \qquad -\infty \leq T_-(z) < 0 < T_+(z) \leq \infty.$$

Every solution through z is a restriction of x_z. If $T_+(z) < \infty$, then for every compact $K \subset U$ there is a time $t_K < T_+(z)$ such that $x_z(t) \notin K$ for all $t_K < t < T_+(z)$. The corresponding conclusion holds at a finite $T_-(z)$ as time decreases.

Proof. Take the union of the intervals of all solutions through z. It is an open interval containing zero. On overlaps the solutions agree by Theorem 2.1, so their values define one function on this union. Near every time this function agrees with a solution. It is therefore a solution itself, and the construction makes it the unique maximal one.

The escape claim is immediate if K is empty. For nonempty K, each $a \in K$ has, by Theorem 2.1, an open neighbourhood V_a of initial states and a common time $\tau_a > 0$. Finitely many of these neighbourhoods cover K. Their minimum time $\tau > 0$ works for every initial state in K. Suppose $T_+(z)$ is finite but the solution does not eventually leave K. Then there are times $t_j \uparrow T_+(z)$ with $x_z(t_j) \in K$. The local solution starting at $x_z(t_j)$ exists for τ units of time in both directions. Shift it to start at t_j. Uniqueness identifies it with x_z on their overlap. For j large enough, $t_j + \tau > T_+(z)$, so gluing them extends the maximal solution. This is a contradiction. Apply the same argument to $-F$ for the negative endpoint. ∎

For $\dot{x} = x^2$ and $x(0) = 1$, direct differentiation verifies

$$x(t) = \frac{1}{1 - t}, \qquad J_1 = (-\infty, 1).$$

The value tends to infinity at the finite endpoint. For $\dot{x} = 1$ on $U = (0, 1)$, with $x(0) = 1/2$, the maximal solution is $x(t) = 1/2 + t$ on $(-1/2, 1/2)$. It stays bounded but tends to the boundary of U. Compact escape includes both behaviours.

2.5 The flow and its derivatives

The *flow* records the maximal solution as a function of its initial state. Write $\varphi_t(z) = x_z(t)$, always retaining the time subscript. The *domain of the flow* is

$$W = \{(t, z) \in \mathbb{R} \times U : T_-(z) < t < T_+(z)\}.$$

Here domain means the set on which this map is defined; connectedness is not assumed. A homeomorphism is a continuous bijection with continuous inverse.

Theorem 2.5 (Continuous local flow). *For $s \in J_z$, the shifted interval is $J_{\varphi_s(z)} = J_z - s$, and*

$$\varphi_0(z) = z, \qquad \varphi_t(\varphi_s(z)) = \varphi_{t+s}(z) \quad (s, t+s \in J_z). \tag{2.4}$$

The set W is open, and $(t, z) \mapsto \varphi_t(z)$ is continuous on W. For fixed t, the map φ_t is a homeomorphism from $W_t = \{z \in U : (t, z) \in W\}$ onto W_{-t}, with inverse φ_{-t}.

Proof. Time translation preserves the equation because it is autonomous. The path $t \mapsto x_z(s + t)$ is a solution through $\varphi_s(z)$ on $J_z - s$. It is maximal: an extension, shifted back, would extend x_z. Uniqueness proves the interval identity and the group law.

The local construction and Equation (2.2) already give a continuous flow on $(-\tau_a, \tau_a) \times V_a$ around each $(0, a)$. Fix $(t_0, z_0) \in W$. The reference path between zero and t_0 is compact. Finitely many of the sets V_a cover its image. The minimum of their times is positive. Thus there is an open set O containing that image and a number $h > 0$ such that the local flow is defined and continuous on $(-h, h) \times O$.

Choose an integer N with $|t_0| / N < h$ and let $Q : O \to U$ be given by $Q(y) = \varphi_{t_0/N}(y)$. Each reference state $Q^j(z_0)$, for $0 \le j \le N$, lies in O. By continuity, the finitely many compositions are defined on an open neighbourhood V of z_0, and $Q^N(V) \subset O$ after shrinking V. Consequently,

$$\varphi_t(z) = \varphi_{t-t_0}(Q^N(z)), \qquad |t - t_0| < h, \quad z \in V.$$

The right-hand side constructs a solution at these times by gluing local solutions. By Theorem 2.4, it is a restriction of the maximal solution. It is continuous, and its rectangle lies in W. This proves openness and continuity, including $t_0 = 0$ by the initial local construction. Finally, the group law with times t and $-t$ proves that φ_t and φ_{-t} are inverse maps between W_t and W_{-t}. These sets are open because W is open. ∎

For $\dot{x} = 1$ on $(0, 1)$, the flow is $\varphi_t(z) = z + t$ and its domain is $W = \{(t, z) : 0 < z < 1,\ -z < t < 1 - z\}$. The time maps compose only while the intermediate and final states remain in $(0, 1)$. An autonomous equation need not give maps on all of U for every time.

The flow also gives dependence over a whole compact time interval.

Corollary 2.6. *Fix $z \in U$ and a compact interval $J \subset J_z$ containing zero. There are a neighbourhood V of z and a constant C such that $J \times V \subset W$ and*

$$\sup_{t \in J} \|\varphi_t(y) - \varphi_t(w)\| \le C \|y - w\|, \qquad y, w \in V.$$

Proof. Each local solution map in the preceding construction is Lipschitz in its initial state by Equation (2.2), uniformly in its short time. For each reference intermediate state choose one local neighbourhood V_{a_j}. Shrink the initial neighbourhood so that both jth intermediate states lie in that same set. A finite composition is Lipschitz, with constant the product of these local constants. The final variable-time step has a uniform local constant too. This

proves the claimed estimate on a time neighbourhood of each $t \in J$. Finitely many cover J. Intersect their initial neighbourhoods and take the maximum of their constants. The flow rectangles give $J \times V \subset W$. ∎

Theorem 2.7 (Smooth dependence · *sketched*). *If $F \in C^k(U, \mathbb{R}^m)$, for an integer $k \geq 1$, then the flow is C^k in (t, z) on W. For fixed z, the variational matrix $M(t) = D_z\varphi_t(z)$ satisfies*

$$\dot{M}(t) = DF(\varphi_t(z))M(t), \qquad M(0) = I. \tag{2.5}$$

Here I is the $m \times m$ identity matrix. Each $\varphi_t : W_t \to W_{-t}$ is a C^k diffeomorphism: a bijection of class C^k with inverse of class C^k.

Sketch. Use the local construction with $a = z$, $0 < H < \tau$ and $h \in \mathbb{R}^m$ small enough that $z + h \in V$. The paths from z and $z + h$ stay in its ball on $[-H, H]$, where F has Lipschitz constant L. Solve the linear integral equation $M(t) = I + \int_0^t DF(\varphi_s(z))M(s)\,\mathrm{d}s$ by contraction on continuous matrix-valued functions. Its coefficient is continuous and bounded there. Put $e_h(t) = \varphi_t(z + h) - \varphi_t(z) - M(t)h$. The separation bound makes the path displacement at most a constant times $\|h\|$. Uniform continuity of DF on a compact neighbourhood of the reference path gives a Taylor remainder bounded by $\varepsilon(\|h\|)\,\|h\|$, where $\varepsilon(r) \to 0$ as $r \downarrow 0$. Subtracting the integral equations therefore gives, for $0 \leq t \leq H$,

$$\|e_h(t)\| \leq L \int_0^t \|e_h(s)\|\,\mathrm{d}s + H\varepsilon(\|h\|)\,\|h\|\,.$$

Grönwall proves $\sup_{0 \leq t \leq H} \|e_h(t)\| / \|h\| \to 0$. Negative times use $-F$. This identifies the derivative with M. The integral equation gives continuous dependence of M on (t, z). The identity $\partial_t \varphi_t(z) = F(\varphi_t(z))$ supplies time derivatives. Higher initial-state derivatives satisfy linear equations in the highest unknown derivative, with known terms involving lower derivatives. The omitted steps are the uniform remainder estimates and induction for these higher and mixed derivatives. Finite composition extends the local conclusion over W. The inverse is φ_{-t} by the proved group law.

For example, for $F(x) = \lambda x$ the variational matrix is the scalar $M(t) = e^{\lambda t}$, which solves $\dot{M} = \lambda M$ and $M(0) = 1$.

2.6 Global existence and the pendulum

A solution is *global* if its interval is all of $\mathbb{R}$. A field is *complete* if every maximal solution is global. Here completeness concerns time of existence, rather than completeness of a normed space. A growth bound gives one sufficient condition.

Proposition 2.8 (At most linear growth). *Let $F \in C^1(\mathbb{R}^m, \mathbb{R}^m)$ satisfy $\|F(x)\| \leq a + b\,\|x\|$ for all $x \in \mathbb{R}^m$, where $a, b \geq 0$. Then F is complete.*

Proof. Fix $H > 0$. For $0 \leq t \leq H$ in the solution's interval, the integral equation and Grönwall give

$$\|x_z(t)\| \leq \|z\| + aH + b \int_0^t \|x_z(s)\|\,\mathrm{d}s \leq (\|z\| + aH)e^{bH}.$$

If $T_+(z)$ were finite, choose $H > T_+(z)$. The bound traps the solution in a compact ball up to that endpoint, contradicting Theorem 2.4. The same estimate for $-F$ excludes a finite negative endpoint. ∎

In particular, constant-coefficient systems $\dot{x} = Ax$ are global because $\|Ax\| \le \|A\| \, \|x\|$.

More generally, let $A : \mathbb{R} \to \mathbb{R}^{m \times m}$ and $f : \mathbb{R} \to \mathbb{R}^m$ be continuous. The linear system $\dot{x} = A(t)x + f(t)$ has a unique global solution through each initial state at time zero. Here a solution is a C^1 path satisfying that equation and its initial value. Fix $H > 0$ and let $L_H = \sup_{|t| \le H} \|A(t)\|$. The integral map is a contraction on the whole continuous path space on each step whenever its length times L_H is less than one. It maps that space to itself since its coefficients are continuous. Finitely many steps cover each bounded interval in either direction, proving the claim.

For the pendulum, recall $\dot{x} = v$ and $\dot{v} = -\nu_0^2 \sin x$, with $\nu_0 > 0$ the small-oscillation frequency. Its energy is

$$E(x, v) = \frac{v^2}{2} + \nu_0^2 (1 - \cos x).$$

Along a solution the chain rule gives $\dot{E} = v(-\nu_0^2 \sin x) + \nu_0^2 \sin x \, v = 0$. For initial state (x_0, v_0), write $E_0 = E(x_0, v_0)$. Nonnegativity of the potential gives

$$|v(t)| \le \sqrt{2E_0}, \qquad |x(t) - x_0| \le \sqrt{2E_0}\, |t| \,.$$

For a putative finite positive endpoint T_+, the states stay in the compact rectangle $|x - x_0| \le \sqrt{2E_0} T_+$ and $|v| \le \sqrt{2E_0}$, contained in $\mathbb{R}^2$. Compact escape rules out this endpoint. Reverse time to rule out a finite negative endpoint. Thus every pendulum motion exists for all time, including equilibria and motions whose angle is unbounded as time increases.

Summary.

- A C^1 autonomous field on an open set has a unique maximal solution through each state. At a finite endpoint it eventually leaves every compact subset of that open set.
- A Lipschitz constant L on a common region gives the separation bound $e^{L|t|} \, \|z - w\|$ while both paths remain there.
- The flow has an open domain and satisfies the group law whenever the indicated times belong to its maximal intervals. Smooth dependence and the variational equation were sketched.
- At most linear growth on all of $\mathbb{R}^m$ ensures global existence. The pendulum is global by its energy bound.

These results justify the solution and flow assumptions used in mechanics.

References

G. Teschl, *Ordinary Differential Equations and Dynamical Systems*, AMS GSM 140, 2012, §§2.2–2.6 and Theorem 6.1. R. S. Strichartz, *The Way of Analysis*, revised edition, 2000, Chapter 11.